

Extra characters in a string

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Problem Statement

There is a string, and a dictionary.

I need to break string optimally such that each substring is in dictionary.

o/p: min number of extra characters left over.

Example

Eg: aadyazzrastogi

dictionary = { aadya, rastogi }

hence string can be broke to aadya, rastogi

In between (zz) → unused → **o/p = 2**

Approach

$dp[i]$ = min extra characters in first i characters

Consider the split: $[0 \dots j-1]$ $[j \dots i-1]$

- now if $s[j \dots i-1]$ is in dictionary
→ add 0 as no extra
- if not in dictionary
→ add $(i-1) - j + 1 = (i-j)$

Code Implementation

```
class Solution {
public:
    int minExtraChar(string s, vector<string>& dictionary) {
        int n=s.size();

        //vasic idea yahi rhega ki
        //[0.....j-1][j.....i-1]

        //agar substring (j...i-1) dictionary me he to no extra char
        //agar dictionary me nhi he to i-j extra characters

        unordered_set<string> st;
        for(auto wrd:dictionary){
            st.insert(wrd);
        }

        //for o(1) lookup

        vector<int> dp(n+1,INT_MAX);
        dp[0]=0;

        for(int i=1;i<=n;i++){
            for(int j=i-1;j>=0;j--){

                string word=s.substr(j,i-j);
                if(st.find(word)!=st.end()){
                    dp[i]=min(dp[i],dp[j]+0);
                }

                else{
                    dp[i]=min(dp[i],dp[j]+(i-j));
                }
            }
        }
        return dp[n];
    }
};
```